

CS 101: Assignment 1 - Python Fundamentals

Department of Computer Science | Maximum Marks: 100

Problem 1: Control Flow & Conditionals (30 Marks)

Implement a command-line utility in Python that reads student grades from an input string, computes weighted GPA, and categorizes performance according to the institutional grading scale.

Problem 2: Recursion & Memoization (35 Marks)

Write a recursive solution to compute the N-th Fibonacci sequence value with top-down memoization. Compare execution time between standard recursion and memoized calls.

Problem 3: File I/O & Error Handling (35 Marks)

Design a script that reads a CSV dataset of server access logs, parses HTTP status codes, and outputs an error frequency report while gracefully handling missing files.